Phylogenetic Tree Analysis — Cheat Sheet

Hand-in 1 background theory · distances, substitution models, tree methods, bootstrap, coding-sequence evolution, model selection

3 Tree-building methods

Method	Basis	Key assumption	Watch out for
UPGMA	distance, clustering	strict molecular clock (equal rates)	wrong topology if rates vary; only method that's inherently rooted
NJ	distance, min. evolution	additive distances, no clock	fast, single tree, no likelihood score
MP	character-based	fewest total changes wins	long-branch attraction; often several equally parsimonious trees; ignores multiple hits
ML	character-based, model	explicit substitution model	slow; heuristic search (NNI/SPR); depends on starting tree

MP & ML search tree space heuristically — record your search settings.

1 Alignment & data prep

Alignment defines **homology**: every column is a claim that those residues share a common ancestral position. A bad alignment corrupts every downstream calculation.

- **ClustalW** (progressive, guide-tree based) vs **MUSCLE** (iterative refinement) — state which you used.
- **Gap handling:** complete deletion (drop any column with a gap) / pairwise deletion / partial deletion with a site-coverage cutoff — changes the number of sites analysed → changes distances.
- **Coding sequences** (whales, BRCA1, EKAF): given "in reading frame" means position 1 is the start of a codon, so every 3 bases = one amino acid with no leftover partial codon. Align these with a **codon-aware method** (MEGA: Alignment Explorer → Align Codons, or align the translated protein and back-translate) so gaps are only ever inserted in multiples of 3. A plain nucleotide alignment can insert a 1- or 2-base gap and shift the frame for every codon after it.

If a dataset is already aligned, build the tree directly — don't re-unalign/re-align it (or say explicitly that you did).

2 Distances & substitution models

p-distance = raw proportion of differing sites. It **saturates** over time: multiple hits / back-mutations / parallel changes at the same site hide real substitutions → p-distance *underestimates* true divergence.

Model-corrected distance rescales p into an estimate of the true number of substitutions/site — always ≥ p, gap widens with divergence. **This is why K2P branch lengths > p-distance branch lengths.**

Model hierarchy — each step fixes an unrealistic assumption

JC69: assumes every substitution (any base→any other) is equally likely.
K2P: adds **R**, the transition/transversion rate ratio, as an explicit parameter — because transitions (purine↔purine, pyrimidine↔pyrimidine) are chemically easier and happen more often than transversions, so R > 0.5 almost always (mtDNA is especially Ts-biased). **The "Ts/Tv bias" isn't a separate correction on top of the model — it IS the extra parameter K2P estimates**; JC69 ignores it (implicitly assumes R=0.5), so JC69 underestimates divergence on Ts-biased data even more than K2P does.
Tamura-3/HKY/Tamura-Nei: also allow unequal base frequencies (e.g. GC-content bias).
GTR: all 6 pairwise rates free — most parameter-rich.

Rate variation across sites: +G and +I

Not every site evolves at the same rate — e.g. 3rd codon positions change far more than structurally critical amino acid sites. Ignoring this **underestimates distance**, because multiple hits piling up at a few fast sites get undercounted.

- **+G** (gamma, shape α, estimated by ML): models that unevenness. Small α (<1) → L-shaped distribution: most sites nearly frozen, a handful of "hot" sites absorb most of the change. Large α (>>1) → distribution flattens toward uniform rates (JC69/K2P's implicit assumption).
- **+I:** a separate proportion of sites assumed to *never* change — removed from the gamma distribution entirely, rather than just given a very low rate.

Model selection in MEGA: Models → Find Best DNA/Protein Model, ranked by **BIC** (or AICc, lower is better) — this is how you get R, α and I for 2.1.

5 Bootstrap

Procedure: resample alignment *columns* with replacement to the original alignment length → rebuild the tree → repeat N times (e.g. 1000) → for each internal branch, count the % of replicate trees recovering that exact bipartition (split of taxa into two groups).

A bootstrap value belongs to an internal branch/split, not a taxon — terminal branches have no bootstrap value.

Interpretation: measures repeatability of the signal given this alignment & method — *not* the probability the clade is true. Systematic error (wrong model, bad alignment, LBA) can give high support to a wrong clade.

- >70% commonly called "significant"; >80–95% strong support.
- Low values → short internal branches / too few informative sites.
- Values differ between methods because each uses different information (all sites vs. only parsimony-informative sites) — not directly comparable across methods.

4 Rooting & topology

Unrooted tree: shows only the branching pattern — who is closer to whom — with no arrow of time and no stated ancestor. This is literally what NJ, MP and ML compute: an unrooted network of relationships.

Rooted tree: adds a starting point (the common ancestor), letting you read the order of divergence, e.g. "X split off before Y and Z separated." Rooting is an assumption added *on top of* the tree-building method — except UPGMA, which comes out rooted automatically because its clock assumption fixes the root equidistant from every tip.

Outgroup rooting: add a taxon known, from independent evidence, to have diverged before all the others (e.g. mouse relative to the primates), then root the unrooted tree on the branch leading to it. **MEGA draws some root by default even with no outgroup chosen — don't mistake that default placement for a real result.**

Topology = the branching pattern only, e.g. for 3 taxa, ((A,B),C) and (A,(B,C)) are *different* topologies. **Branch length** = the amount of change along a branch. Two trees can share a topology but disagree on lengths (different model — see 1.2), or the reverse — say explicitly which one you're comparing.

Resolution: a fully resolved tree has a distinct split at every internal node with decent bootstrap support. Short internal branches / near-polytomies / low bootstrap mean the data can't confidently tell you the order the lineages split in.

8 AIC vs BIC (model selection)

Both balance **fit** (log-likelihood, L) against **complexity** (number of parameters, k) so a model can't win purely by having more free parameters.

$AIC = 2k - 2\ln(L) \quad BIC = k \cdot \ln(n) - 2\ln(L)$

n = number of sites. Since ln(n) > 2 for any alignment longer than ~7 sites, **BIC penalises extra parameters harder than AIC** — so BIC tends to favour simpler models than AIC, especially on long alignments. **AICc** is AIC with a small-sample correction (what MEGA reports as "AIC").

Lower value = better trade-off for either. MEGA's Find Best Model ranks candidates by BIC by default — state in your report which criterion you used, since AIC and BIC occasionally disagree on the winner.

6 Coding sequences: syn vs non-syn

- ~75% of **3rd-codon-position** changes are synonymous (degenerate code); almost all 1st/2nd-position changes are not.
- **Synonymous (dS):** near-neutral → accumulates fast → good molecular clock, but **saturates first** on deep branches.
- **Non-synonymous (dN):** usually under **purifying selection** → changes slowly, conservatively.
- **dN/dS < 1** = purifying selection (constraint); **dN/dS > 1** = positive/diversifying selection.

In MEGA: Substitution Type → "Syn-Nonsynonymous" when building the tree to split site classes.

Protein trees (translated, aligned as protein) only "see" non-synonymous signal → should resemble the non-synonymous-site tree. Protein distance models: Poisson, Dayhoff/PAM, JTT. Use the correct genetic code table (standard vs. vertebrate mitochondrial).

7 Gene trees vs species trees

mtDNA = a single, maternally-inherited, non-recombining locus → one **gene tree**, which need not match the true **species tree**, due to:

- **Incomplete lineage sorting (ILS):** right before a species splits, its ancestral population already carries several co-existing gene variants. Which variant a given descendant ends up with is down to chance in that population (genetic drift) — not to the true order the species split in. If two splitting events happen close together in time (short internal branches), a lineage can by chance keep a variant that groups it with the "wrong" relative — producing a gene tree that disagrees with the species tree even with *zero* gene flow.
- **Introgression / admixture** — actual interbreeding between lineages after they'd started to diverge.
- Ordinary stochastic variance in the coalescent process, on top of ILS itself.

Relevant for homomtDNA: nuclear DNA gives species splits (human vs Nea+Den 700–500 kya; Nea vs Den 400–300 kya). If your mtDNA tree conflicts with this, name the phenomenon and use bootstrap support to judge whether the conflict is real signal or noise.

9 Testing your assumptions

- **Molecular clock test** (MEGA: Tests → Molecular Clock Test): a likelihood-ratio test comparing a tree forced to be clock-like against the unconstrained ML tree. A significant result means rates differ across lineages — don't assume a clock (UPGMA, or EKAF 5.1's constant-rate claim) without running this first.
- **Detecting saturation:** plot transitions/transversions (or p-distance) against model-corrected distance across all pairs — if it flattens or bends over for the most divergent pairs, those sites are saturated and even the "corrected" distance underestimates true divergence. Deep splits (whales, BRCA1) are most at risk.
- **Long-branch attraction (LBA):** two unrelated long branches can be pulled together as false sisters by chance similarity from independent multiple hits — worse under MP (no correction for multiple hits) or an under-parameterised model, and worse with sparse taxon sampling. A tree pairing two long branches while everything else is short is a red flag; cross-check against ML/NJ with a proper model.

★ Dataset quick-reference

Dataset	Focus
Primates	Reproduce NJ/UPGMA (Nei & Kumar); p vs K2P; bootstrap meaning; MP tree.
HomomtDNA	Best-fit model; Ts/Tv; α; NJ + 1000 bootstrap; gene-tree/species-tree discordance.
Whales	UPGMA/NJ/ML/MP compared; topology & bootstrap agreement across methods.
BRCA1	Two methods compared; bootstrap significance (>70%); resolution (>80%).
EKAF	Constant-rate protein → clock method; syn vs non-syn trees; protein tree comparison.

✓ Report checklist — document for every tree

- Alignment method & gap treatment
- Number of sites analysed
- Tree method (UPGMA / NJ / MP / ML)
- Substitution model (+G/+I, parameters where asked)
- Bootstrap replicate count
- Software version (MEGA11/X vs MEGA12)
- Outgroup / rooting choice